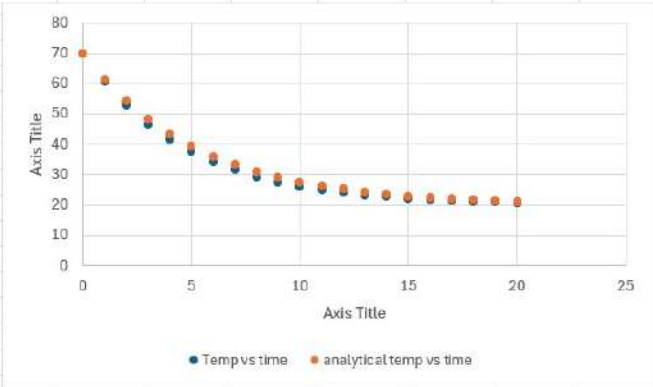
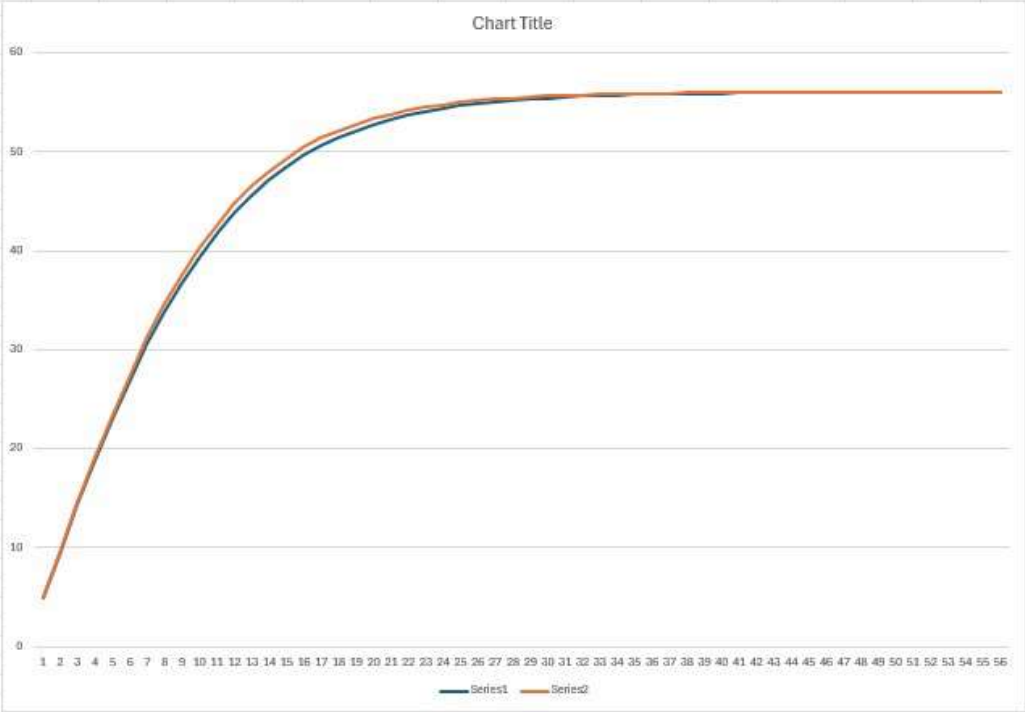
Ti	70
Tf	20
del t	1
K	0.19

time	numerical	dt	analytical
0	70	0	70
1	60.5	1	61.348
2	52.805	1	54.1931
3	46.5721	1	48.2763
4	41.5234	1	43.3833
5	37.4339	1	39.3371
6	34.1215	1	35.991
7	31.4384	1	33.2239
8	29.2651	1	30.9356
9	27.5047	1	29.0433
10	26.0788	1	27.4784
11	24.9239	1	26.1844
12	23.9883	1	25.1142
13	23.2305	1	24.2292
14	22.6167	1	23.4974
15	22.1196	1	22.8922
16	21.7168	1	22.3917
17	21.3906	1	21.9779
18	21.1264	1	21.6356
19	20.9124	1	21.3526
20	20.739	1	21.1185



Cd	0.25
g	9.81
m	80

Time	V(new)	V(old)	Dt	Slope	
0		0	0.5		0
0.5	4.905	4.905	0.5	4.8925	4.905
1	9.7724	9.7724	0.5	9.711	9.7724
1.5	14.528	14.528	0.5	14.386	14.528
2	19.103	19.103	0.5	18.856	19.103
2.5	23.438	23.438	0.5	23.07	23.438
3	27.485	27.485	0.5	26.992	27.485
3.5	31.209	31.209	0.5	30.597	31.209
4	34.593	34.593	0.5	33.875	34.593
4.5	37.628	37.628	0.5	36.823	37.628
5	40.321	40.321	0.5	39.451	40.321
5.5	42.685	42.685	0.5	41.775	42.685
6	44.743	44.743	0.5	43.815	44.743
6.5	46.52	46.52	0.5	45.594	46.52
7	48.044	48.044	0.5	47.137	48.044
7.5	49.342	49.342	0.5	48.469	49.342
8	50.443	50.443	0.5	49.614	50.443
8.5	51.372	51.372	0.5	50.594	51.372
9	52.154		0.5	51.431	52.154
9.5	52.809	52.154	0.5	52.144	52.809
10	53.356	52.809	0.5	52.75	53.356
10.5	53.813	53.356	0.5	53.263	53.813
11	54.193	53.813	0.5	53.698	54.193
11.5	54.509	54.193	0.5	54.066	54.509
12	54.772	54.509	0.5	54.377	54.772
12.5	54.989	54.772	0.5	54.639	54.989
13	55.17	54.989	0.5	54.86	55.17
13.5	55.319	55.17	0.5	55.046	55.319
14	55.442	55.319	0.5	55.202	55.442
14.5	55.544	55.442	0.5	55.334	55.544
15	55.629	55.544	0.5	55.445	55.629
15.5	55.699	55.629	0.5	55.538	55.699
16	55.756	55.699	0.5	55.617	55.756
16.5	55.804	55.756	0.5	55.683	55.804
17	55.843	55.804	0.5	55.738	55.843
17.5	55.875	55.843	0.5	55.785	55.875
18	55.902	55.875	0.5	55.824	55.902
18.5	55.924	55.902	0.5	55.857	55.924
19	55.943	55.924	0.5	55.884	55.943
19.5	55.958	55.943	0.5	55.907	55.958
20	55.97	55.958	0.5	55.927	55.97
20.5	55.98	55.97	0.5	55.943	55.98
21	55.989	55.98	0.5	55.957	55.989
21.5	55.996	55.989	0.5	55.968	55.996
22	56.001	55.996	0.5	55.978	56.001
22.5	56.006	56.001	0.5	55.986	56.006
23	56.01	56.006	0.5	55.993	56.01
23.5	56.013	56.01	0.5	55.999	56.013
24	56.016	56.013	0.5	56.003	56.016
24.5	56.018	56.016	0.5	56.008	56.018
25	56.02	56.018	0.5	56.011	56.02
25.5	56.022	56.02	0.5	56.014	56.022
26	56.023	56.022	0.5	56.016	56.023



→ sample calculation:

$$\frac{dT}{dt} = -k(T - T_\alpha)$$

$$\therefore \int \frac{1}{T - T_\alpha} \cdot dt = \int -k \cdot dt$$ analytical

$$\therefore \ln(T - T_\alpha) = -kt + c$$

$$T - T_\alpha = e^{-kt} \cdot e^{c}$$

$$T - T_\alpha = x e^{-kt}$$

at $t = 0$, $T = T_\alpha + x e^{-kt}$

$$\therefore x = T - T_\alpha$$

$$\therefore T(t) = T_\alpha + (T - T_\alpha) e^{-kt}$$

Euler.

$$T_{n+1} = T_n + \Delta t \frac{dT}{dt}$$

$$T_{n+1} = T_n + \Delta t(-k(T_n - T_\alpha))$$

$$\therefore T_{n+1} = T_n + \Delta t(-k(T_n - T_\alpha))$$

$$\therefore T_{n+1} = T_n - kh(T_n - T_\alpha)$$

$(h = \Delta t)$

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Bunjee jumping:

given : v = velocity , $g = 9.81\ m/s$, $C_d = 0.25$, $m = 68.1$
t = time.

$$\frac{dv}{dt} = g - \frac{C_d}{m^2} v^2$$

$$V_t = \sqrt{\frac{gm}{C_d}} \cdot \tanh\left(\sqrt{\frac{gC_d}{m}} \cdot t\right) \rightarrow \text{analytical eq}^n$$

$$V_{t_2} = V_{t_1} + \left[g - \frac{C_d}{m} V_{t_1}^2\right](t_{i+1} - t_i)$$

↳ numerical eq^n.